



CONFLICT TO COMPLIANCE

A MULTI-AGENT RISC-V EXTENSION
MIGRATION ACROSS **SPEC, HW & SW**

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One non-compliant RISC-V encoding blocks upstreaming and forces spec, RTL, DV, toolchain, emulator, runtime and firmware to move together, wasting time and money.

We propose a **multi-agent migration tool** that leverages **UDB** as the source of truth and an agent setup with a planner and HW/SW/DV specialist agents to implement the changes, keeping architectural decisions to humans.

To prove it, we port two non-compliant packed-SIMD extensions (**PS + PI**) from spec to silicon.

01 THE PROBLEM — ONE ENCODING, THE WHOLE STACK

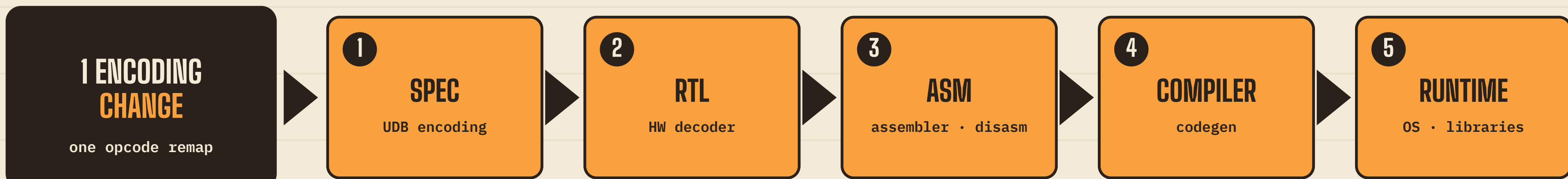
Extensions designed before or outside current allocation practice end up **non-compliant**: they squat on standard or reserved major opcodes.

Custom extensions also face a **packing problem** — only four custom opcodes exist, so every change in a platform's configuration forces a new opcode layout.

RISC-V's **upstream-first** model amplifies the penalty: non-compliant encodings cannot be upstreamed, so users carry the maintenance burden alone and the ecosystem loses reusable functionality.

CAN'T BE UPSTREAMED
locked out of the shared ecosystem

EVERY RE-SPECIFICATION **PROPAGATES THROUGH THE FULL STACK** — BY HAND, LAYER BY LAYER



5+ layers touched per opcode move, lots of engineers involved — very costly, and exactly the repetitive work we automate.

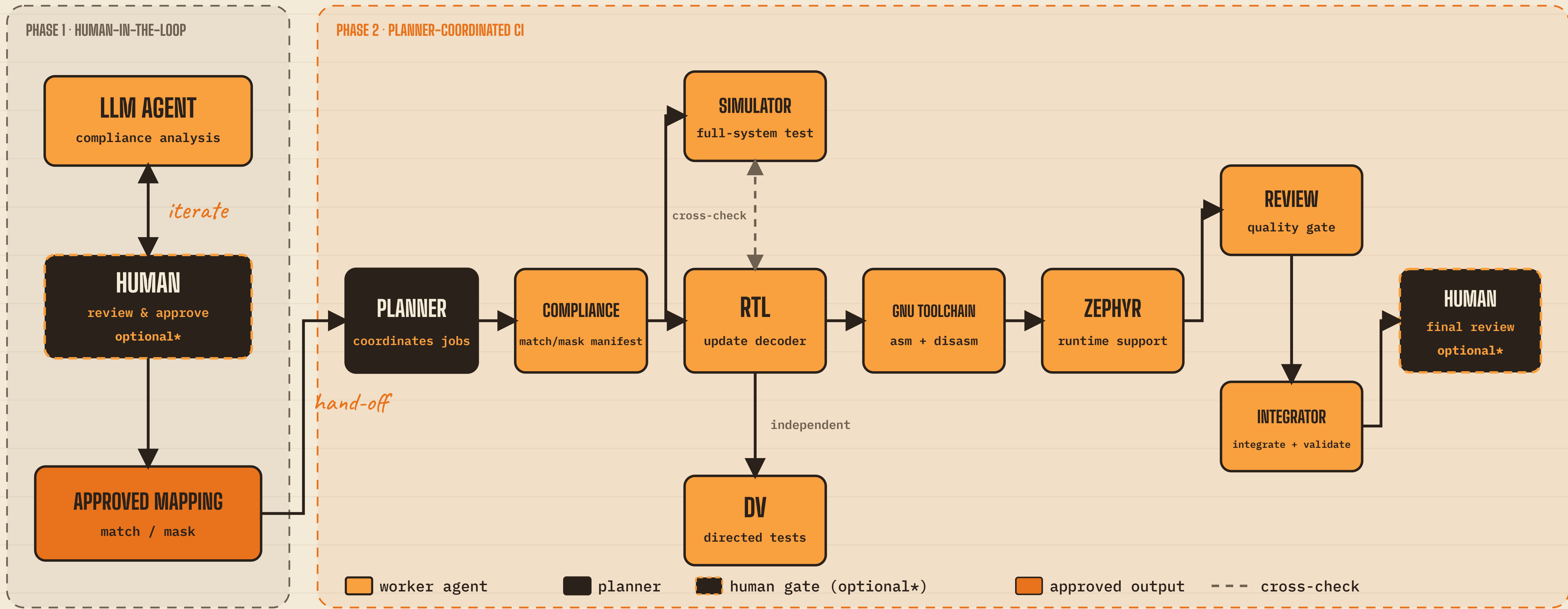
02 THE WORKFLOW — UDB AS TRUTH, LLM AGENTS AS THE ENGINE

PHASE 1 · HUMAN-IN-THE-LOOP

The compliance agent reads encodings from UDB, cross-references ratified & reserved space, and proposes conservative remaps into custom-0...3 — preserving semantics, moving only the bits that must change. In autonomous mode the loop runs end-to-end, but architectural calls are never taken without the architect.

PHASE 2 · PLANNER-COORDINATED CI

The approved mapping flows into the planner, which coordinates the run. Specialised agents — each with its own prompt, tools and model — then generate every layer: the RTL decoder is independently verified by DV (spec-derived tests) and cross-checked against the simulator; review is the quality gate and the integrator assembles & validates the change for an optional final human review.



03 CASE STUDY — PS + PI, SPEC TO SILICON

THE EXTENSIONS

two packed-SIMD ISAs from a taped-out manycore SoC

PS · PACKED-SINGLE 8 × FP32 lanes · 81 ops FP add/mul/FMA, sqrt, exp2 / log2 / recip, compares & masks, FP↔INT & pack/convert, broadcast · gather · swizzle
PI · PACKED-INTEGER 8 × int32 / uint32 · 37 ops add/mul/div, AND/OR/XOR, shifts, signed/unsigned min-max, pack & saturate to 16/8-bit

118 instructions audited — R-type & R4-type
most already live in **custom-0 / 2 / 3**

~10%

SIT ON STANDARD / RESERVED OPCODES

→ 10 **non-compliant** · 8 remapped · 2 escalated

10 NON-COMPLIANT — OLD → COMPLIANT

where each was & where it moved · real match from the manifest · 2 escalated

INSTRUCTION	SITS ON	WAS (OLD)	NOW (CUSTOM) TARGET
f1q2	LOAD-FP	0x00005007	0x0000500b custom-0
fsq2	STORE-FP	0x00005027	0x0000502b custom-1
packb	OP-32	0x8000603b	0x8000602b custom-1
bitmixb	OP-32	0x8000703b	0x8000702b custom-1
faddi.pi	reserved	0x0400003f	0x0400002b custom-1
fandi.pi	reserved	0x0400103f	0x0400102b custom-1
fcmov.ps	reserved	0x0400203f	0x0400202b custom-1
fcmovm.ps	reserved	0x00000077	0x0000002b custom-1
fbci.ps	reserved	0x0000001f	escalated architect
fbci.pi	reserved	0x0000005f	escalated architect

2 escalated. fbci.ps / fbci.pi are U-type (20-bit immediate) — no non-overlapping custom encoding; the architect decides.

GENERATED & PROVEN

across the open stack

UDB riscv-unified-db	✓ match
BINUTILS GNU as / objdump	✓ round-trip
RTL + DV core-et decoder	✓ decode
SYSEMU et-platform	✓ equiv
ZEPHYR runtime	✓ encode
AIFLOW VALIDATE whole set, one command	EXIT 0



MIGRATE YOUR OWN

Pair structured ISA data with tool-driven agents. Architects decide; the agents propagate the change across every layer of the stack.

github.com/AFoliveira/AGENTIC-DEMO

